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2

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MATHEMATICS - 2006

Objective Questions: (Q.1-Q.15)

(1.) $\lim_{n \rightarrow \infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}$ equals

- (a.) 3
- (b.) 2
- (c.) 1
- (d.) 0

(2.) Let $f(x) = (x-2)^{17}(x+5)^{24}$. Then

- (a.) f does not have a critical point at 2
- (b.) f has a minimum at 2
- (c.) f has a maximum at 2
- (d.) f has neither a minimum nor a maximum at 2

(3.) Let $f(x, y) = x^5 y^2 \tan^{-1}\left(\frac{y}{x}\right)$.

Then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ equals

- (a.) $2f$
- (b.) $3f$

(c.) $5f$

(d.) $7f$

- (4.) Let G be the set of all irrational numbers. The interior and the closure of G are denoted by G° and $\bar{G}$, respectively. Then

(a.) $G^\circ = \phi$, $\bar{G} = G$

(b.) $G^\circ = \mathbf{R}$, $\bar{G} = \mathbf{R}$

(c.) $G^\circ = \phi$, $\bar{G} = \mathbf{R}$

(d.) $G^\circ = G$, $\bar{G} = \mathbf{R}$

- (5.) Let $f(x) = \int_{\sin x}^{\cos x} e^{-t^2} dt$.

Then $f'\left(\frac{\pi}{4}\right)$ equals

(a.) $\sqrt{\frac{1}{e}}$

(b.) $-\sqrt{\frac{2}{e}}$

(c.) $\sqrt{\frac{2}{e}}$

(d.) $-\sqrt{\frac{1}{e}}$

- (6.) Let C be the circle taken $x^2 + y^2 = 1$ in the anti-clockwise sense. Then the value of the

integral $\int_C [(2xy^3 + y)dx + (3x^2y^2 + 2x)dy]$

equals

(a.) 1

(b.) $\frac{\pi}{2}$

(c.) π

(d.) 0

(7.) Let r be the distance of a point $P(x, y, z)$ from the origin O . Then ∇r is a vector

(a.) Orthogonal to $\overline{OP}$

(b.) Normal to the level surface of r at P

(c.) Normal to the surface of revolution generated by OP about x -axis

(d.) Normal to the surface of revolution generated by OP about y -axis

(8.) Let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_2, 0).$$

If $N(T)$ and $R(T)$ denote the null space and the range space of T respectively, then

(a.) $\dim N(T) = 2$

(b.) $\dim R(T) = 2$

(c.) $R(T) = N(T)$

(d.) $N(T) \subset R(T)$

- (9.) Let S be a closed surface for which $\int_S \vec{r} \cdot \hat{n} d\sigma = 1$. Then the volume enclosed by the surface is
- (a.) 1
(b.) $\frac{1}{3}$
(c.) $\frac{2}{3}$
(d.) 3
- (10.) If $\frac{(c_1 + c_2 \ln x)}{x}$ is the general solution of the differential equation $x^2 \frac{d^2 y}{dx^2} + kx \frac{dy}{dx} + y = 0$, then k equals
- (a.) 3
(b.) -3
(c.) 2
(d.) -1
- (11.) If A and B are 3×3 real matrices such that $\text{rank}(AB) = 1$, then $\text{rank}(BA)$ cannot be
- (a.) 0
(b.) 1
(c.) 2
(d.) 3

- (12.) The differential equation representing the family of circles touching y -axis at the origin is
- (a.) Linear and of first order
 - (b.) Linear and of second order
 - (c.) Non-linear and of first order
 - (d.) Non-linear and of second order
- (13.) Let G be a group of order 7 and $\phi(x) = x^4$, $x \in G$. Then ϕ is
- (a.) Not one-one
 - (b.) Not onto
 - (c.) Not a homomorphism
 - (d.) One-one, onto and a homomorphism
- (14.) Let R be the ring of all 2×2 matrices with integer entries. Which of the following subsets of R is an integral domain?
- (a.) $\left\{ \begin{pmatrix} 0 & x \\ y & 0 \end{pmatrix} : x, y \in \mathbf{Z} \right\}$
 - (b.) $\left\{ \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix} : x, y \in \mathbf{Z} \right\}$
 - (c.) $\left\{ \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} : x \in \mathbf{Z} \right\}$
 - (d.) $\left\{ \begin{pmatrix} x & y \\ y & z \end{pmatrix} : x, y, z \in \mathbf{Z} \right\}$

(15.) Let $f_n(x) = n \sin^{2n+1} x \cos x$. Then the value of

$$\lim_{n \rightarrow \infty} \int_0^{\pi/2} f_n(x) dx - \int_0^{\pi/2} \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx \text{ is}$$

(a.) $\frac{1}{2}$

(b.) 0

(c.) $-\frac{1}{2}$

(d.) $-\infty$

Subjective Questions: (Q.16-Q.29)

(16.) (a.) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n^n}{n! 3^n}$.

(b.) Show that $\ln(1 + \cos x) \leq \ln 2 - \frac{x^2}{4}$ for

$$0 \leq x \leq \frac{\pi}{2}.$$

(17.) Find the critical points of the function

$$f(x, y) = x^3 + y^2 - 12x - 6y + 40.$$

Test each of these for maximum and minimum.

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(18.) (a.) Evaluate $\iint_R x e^{y^2} dx dy$, where R is the region bounded by the lines $x = 0$, $y = 1$ and the parabola $y = x^2$.

(b.) Find the volume of the solid bounded above by the surface $z = 1 - x^2 - y^2$ and below by the plane $z = 0$.

(19.) Evaluate the surface integral

$$\iint_S x(12y - y^4 + z^2) d\sigma,$$

where the surface S is represented in the form $z = y^2$, $0 \leq x \leq 1$, $0 \leq y \leq 1$.

(20.) Using the change of variables, evaluate $\iint_R xy \, dx \, dy$, where the region R is bounded by the curves $xy = 1$, $xy = 3$, $y = 3x$ and $y = 5x$ in the first quadrant.

(21.) (a.) Let u and v be the eigenvectors of A corresponding to the eigen values 1 and 3 respectively. Prove that $u + v$ is not an eigen vector of A .

(b.) Let A and B be real matrices such that the sum of each row of A is 1 and the sum of each row of B is 2. Then show that 2 is an eigenvalue of AB .

- (22.) Suppose W_1 and W_2 are subspaces of $\mathbf{R}^4$ spanned by $\{(1, 2, 3, 4), (2, 1, 1, 2)\}$ and $\{(1, 0, 1, 0), (3, 0, 1, 0)\}$ respectively. Find a basis of $W_1 \cap W_2$. Also find a basis of $W_1 + W_2$ containing $\{(1, 0, 1, 0), (3, 0, 1, 0)\}$.
- (23.) Determine y_0 such that the solution of the differential equation $y' - y = 1 - e^{-x}$, $y(0) = y_0$ has a finite limit as $x \rightarrow \infty$.
- (24.) Let $\phi(x, y, z) = e^x \sin y$. Evaluate the surface integral $\iint_S \frac{\partial \phi}{\partial n} d\sigma$, where S is the surface of the cube $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$ and $\frac{\partial \phi}{\partial n}$ is the directional derivative of ϕ in the direction of unit outward normal to S . Verify the divergence theorem.
- (25.) Let $y = f(x)$ be a twice continuously differentiable function on $(0, \infty)$ satisfying $f(1) = 1$ and $f'(x) = \frac{1}{2} f\left(\frac{1}{x}\right)$, $x > 0$.
 Form the second order differential equation satisfied by $y = f(x)$, and obtain its solution satisfying the given conditions.

- (26.) Let $G = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbf{Z} \right\}$ be the group under matrix addition and H be the subgroup of G consisting of matrices with even entries. Find the order of the quotient group $\frac{G}{H}$.

- (27.) Let $f(x) = \begin{cases} x^2 & 0 \leq x \leq 1 \\ \sqrt{x} & x > 1 \end{cases}$
 Show that f is uniformly continuous on $[0, \infty)$.

- (28.) Find $M_n = \max_{x \geq 0} \left\{ \frac{x}{n(1+nx^3)} \right\}$, and hence prove that the series $\sum_{n=1}^{\infty} \frac{x}{n(1+nx^3)}$ is uniformly convergent on $[0, \infty)$.

- (29.) Let R be the ring of polynomials with real coefficients under polynomial addition and polynomial multiplication. Suppose $I = \{p \in R : \text{sum of coefficients of } p \text{ is zero}\}$. Prove that I is a maximal ideal of R .

Rajendra Dubey (Known as Dubey Sir): He has experience of almost three decades of teaching mathematics. During his journey, he has guided IIT-JEE, IAS, NET, GATE, IIT-JAM aspirants etc. Apart from producing top rankers in the various exams, he is widely known for making mathematics interesting and fascinating. He is founder & director of DIPS ACADEMY and chairman of Dubey's Group of Companies.



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